Andrew Q. Nguyen

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SUMMARY

CS graduate student with experience in LLM evaluation and deployment, RAG pipeline engineering, and network automation at scale. Skilled in GPU-optimized and edge inference, ML infrastructure, and L2/L3 protocol diagnostics across large-scale WAN environments.

EDUCATION

Northeastern University (GPA: 4.0) [Khoury Align Scholarship Awarded] Seattle, Washington
M.S. *in Computer Science* *Expected Graduation, Dec 2026*

University of California San Diego San Diego, California
B.S. *in Biochemistry and Cell Biology* *Graduated, March 2020*

SKILLS

Programming Languages / Integration: Python (proficient), Java (proficient), C++ (familiar), YAML, Bash, KQL, RESTful APIs, TDD
Networking & ML: OSI Model (L1-L3), TCP/IP, VLANs, DHCP, OSPF/BGP (familiar), ACLs, Databricks, PyTorch, MLflow, Spark, Clustering, Large Language Models (LLM), Inference, Deployment, Evaluation, RAG, AWS Lambda, ONNX
DevOps / Databases: Git, Grafana, Docker, Kubernetes, Jenkins, CI/CD, VMs, KQL Database, Delta Lake, MongoDB, Postgres, Redis, Agile

WORK EXPERIENCES

Centene Corporation Seattle, Washington (Remote)
Machine Learning Engineer Intern *September 2025 – Present*

- **Engineered** an end-to-end **LLM evaluation pipeline** using **MLFlow 3.4** and **Spark SQL** to aggregate performance metrics into **Databricks Unity Catalog**, powering a Streamlit recommender app that optimized selection with projected savings of **~$15k/month**.
- **Deployed** GPT-OSS-20B on 4x A10G GPUs with custom **Pyfunc signatures** for **OpenAI API** compatibility, achieving **50%** latency reduction through GPU memory persistence and **bfloat16** precision.
- **Expanded** app with a **RAG evaluation engine** benchmarking 8 **chunking-embedding configurations** across 2 domains via Recall@K and MRR, reducing stack selection from **days to minutes**.

Microsoft Redmond, Washington
Software Engineer Intern *Jun 2025 – Aug 2025*

- **Automated configuration drift** detection and built a **Grafana** dashboard for **Microsoft's WAN** spanning thousands of network devices, achieving a **25%** reduction in incident response time within an **agile** environment using a **TDD** approach.
- **Developed** an automated **YAML workflow** that gathers metadata using **KQL** from a **KQL database,** generates CR/ICM reports via **API** access, then initiates an internal tool to push configuration changes to devices, effectively eliminating **90%** of manual work.
- **Maintained** networking systems by **conducting** Tier-2 diagnostics across L2/L3 protocols including **TCP/IP**, **DHCP**, and **ACLs**.

Dr. Alex Yao, San Diego State University San Diego, California
Cloud Data Intern *Dec 2023 – June 2024*

- **Optimized** model accuracy by **12%** by focusing on feature selection and hyperparameter tuning in a product recommendation system using **PyTorch** and e-commerce data stored in **MongoDB**.
- **Enhanced** system scalability and performance by deploying machine learning models via **AWS Lambda** and **Kubernetes**, integrating with **Postgres** and leveraging **Redis** for in-memory caching to optimize data retrieval.
- **Streamlined** model deployment by setting up **CI/CD** pipelines with **Jenkins** and **Docker**, ensuring scalable releases w/minimal errors.

SELECTED PROJECTS

ConquestFour - Qualcomm & Microsoft On-Device AI Hackathon [Demo] Seattle, Washington
Team of 5 Co-Lead Developer *March 2025*

- **Won Second Place out of 28 Teams** creating a **local LLM**-powered Connect Four game using **Mistral-7B (**4-bit quantized).
- **Implemented** Minimax algorithm with Alpha-Beta pruning and **Z3 state validation**, integrated with speech-to-text capabilities using **OpenAI Whisper**, increasing overall player-AI interaction by **75%**.
- **Optimized** inference by converting PyTorch models to **ONNX Runtime** (INT4 quantization) for **NPU execution**, reducing delay by **60%**.

Semantic Sounds – A Personalized Recommender [Demo] Seattle, Washington
Team of 3 Lead Developer *Dec 2024*

- **Designed** a semantic meaning music recommender system with improved relevancy and **HDBSCAN clustering** effectiveness (Silhouette score: **0.7464**) from base recommender using SHAP-selected features and **sBERT** embeddings.
- **Preprocessed** 60,000 Spotify entries achieving regression models' accuracies of **>.5** to identify features influencing song popularity.

CERTIFICATIONS AND ACTIVITIES

Certifications Online
Google Cyber Security Certificate (Linux, MySQL, and Python hands-on labs) *Completed Nov 2024*
Azure Fundamentals (AZ-900) *Completed June 2025*

Selected for Graduate Leadership Institute at Northeastern (GLI) | *Leadership Development* | *Dec 2024* Seattle, Washington

- **Developed** leadership skills through workshops and feedback sessions, resulting in 40% improvement in team performance.